LAB 3: ADVANCED GRC PHISHING SIMULATION & QUANTITATIVE RISK ANALYSIS

GlobalTech Inc. | Governance, Risk, and Compliance (GRC) Laboratory Deliverable

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DELIVERABLE 1: COMPLETED CALCULATION WORKSHEETS (PHASES 1 & 2)

Phase 1, Step 1: Fundamental Rates

Formula: Rate = (Number of Events / Number of Emails Sent) × 100%

CAMPAIGN	EMAIL OPEN RATE	CLICK-THROUGH RATE	CREDENTIAL SUBMISSION RATE	DATA ENTRY RATE
Campaign A: HR Bonus	(82/100) × 100 = **82.0%**	(47/100) × 100 = **47.0%**	(18/100) × 100 = **18.0%**	(5/100) × 100 = **5.0%**
Campaign B: IT Password Reset	(200/250) × 100 = **80.0%**	(150/250) × 100 = **60.0%**	(75/250) × 100 = **30.0%**	(25/250) × 100 = **10.0%**
Campaign C: CEO Urgent Doc	(72/80) × 100 = **90.0%**	(60/80) × 100 = **75.0%**	(12/80) × 100 = **15.0%**	(8/80) × 100 = **10.0%**

Phase 1, Step 2: Efficiency Metrics

CAMPAIGN	CONVERSION RATE (OPEN → CLICK)	POST-CLICK CREDENTIAL SUBMISSION RATE
Campaign A	(47 / 82) × 100 = **57.3%**	(18 / 47) × 100 = **38.3%**
Campaign B	(150 / 200) × 100 = **75.0%**	(75 / 150) × 100 = **50.0%**
Campaign C	(60 / 72) × 100 = **83.3%**	(12 / 60) × 100 = **20.0%**

Phase 2, Step 3 & 4: Organizational Exposure & Financial Impact Analysis

Baseline Assumptions: Total Organization = 2,000 Employees | Cost per Data Breach = $4,350,000 | Breach Probability upon Credential Theft = 25%

METRIC / STEP	CAMPAIGN B CALCULATION (BASELINE / PEAK RISK)	RESULT
Expected Credentials Stolen	2,000 employees × (75 / 250) = 2,000 × 0.30	**600 employees**
Expected Data Breaches	600 compromised employees × 25% breach probability	**150 breaches**
Total Potential Loss Exposure	150 expected breaches × $4,350,000 per breach	**$652,500,000**
Annualized Loss Expectancy (ALE)	SLE ($4,350,000) × ARO (150 expected breaches)	**$652,500,000**

Phase 2, Step 5: Control Effectiveness & Cost-Benefit Analysis

- **Multi-Factor Authentication (MFA) Implementation:**
  - **Cost:** 2,000 users × $45/user = `$90,000` | **Effectiveness:** 99.9% risk reduction
  - **Risk Reduction Amount:** $652,500,000 × 0.999 = `$651,847,500`
  - **ROI:** ($651,847,500 − $90,000) / $90,000 = **`724,175%`**
- **Security Awareness Training:**
  - **Cost:** 2,000 users × $50/user = `$100,000` | **Effectiveness:** 60% reduction in click-through rates
  - **New Click-Through Rate:** 30% × (1 − 0.60) = `12%`
  - **New Credentials Stolen:** 2,000 × 0.12 = `240 employees`
  - **New Expected Breaches & ALE:** 240 × 0.25 = `60 breaches` → 60 × $4,350,000 = `$261,000,000`
  - **Risk Reduction Amount:** $652,500,000 − $261,000,000 = `$391,500,000`
  - **ROI:** ($391,500,000 − $100,000) / $100,000 = **`391,400%`**

DELIVERABLE 2: RISK ASSESSMENT MATRIX (PHASE 3)

CAMPAIGN	OVERALL RISK SCORE	FINANCIAL EXPOSURE	PRIORITY LEVEL	RECOMMENDED ACTION
Campaign A: HR Bonus	MEDIUM-HIGH	$156,600,000	2	Enhanced Training & Role-Specific Education
Campaign B: IT Password Reset	CRITICAL	$652,500,000	1	IMMEDIATE Enterprise MFA Deployment
Campaign C: CEO Urgent Doc	HIGH	$104,400,000	2	Executive & Assistant Targeted Protocol Training

*Calculations: Campaign A Exposure = (18/100) × 2,000 = 360 employees × 0.25 × $4,350,000 = $156.6M. Campaign C Exposure = (12/80) × 2,000 = 300 employees × 0.25 × $4,350,000 = $104.4M.

DELIVERABLE 3: STATISTICAL ANALYSIS WITH CONFIDENCE INTERVALS

Formula: $p \pm 1.96 \times \sqrt{[p(1-p)/n]}$ (95% Confidence Level)

Campaign B Baseline (Credential Theft Rate = 30%)

- **p:** 0.30 | **n:** 250
- **Standard Error (SE):** √[0.30(0.70) / 250] = √[0.21 / 250] = √0.00084 ≈ 0.02898
- **Margin of Error (MoE):** 1.96 × 0.02898 = **0.0568 (5.7%)**
- **True Rate Range:** 30% ± 5.7% = **24.3% to 35.7%**

Campaign A Student Calculation (Credential Theft Rate = 18%)

- **p:** 0.18 | **n:** 100
- **Standard Error (SE):** √[0.18(0.82) / 100] = √[0.1476 / 100] = √0.001476 ≈ 0.03842
- **Margin of Error (MoE):** 1.96 × 0.03842 = **0.0753 (7.53%)**
- **True Rate Range:** 18% ± 7.53% = **10.47% to 25.53%**

DELIVERABLE 4: EXECUTIVE BRIEFING

TO: Board of Directors, Chief Information Security Officer (CISO), Chief Financial Officer (CFO)

FROM: Chijioke Ezumah, Lead GRC Risk Analyst

DATE: November 1, 2026 | **SUBJECT:** Comprehensive Quantitative Risk Assessment & Security Investment Justification

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1. Executive Summary:

Our recent quarterly phishing simulation series reveals an alarming financial exposure of **$652.5 Million**. The "IT System Password Reset" campaign achieved a 30% credential theft rate, projecting to 600 compromised user accounts across GlobalTech Inc. Without security controls, the expected annual loss from phishing-initiated data breaches threatens organizational viability. Immediate implementation of controls is strongly advised.

2. Key Quantitative Findings:

RISK METRIC	GLOBALTECH VALUE	INDUSTRY AVERAGE	SEVERITY LEVEL
Overall Credential Theft Rate	**22.1%** (105 / 430)	15.3%	HIGH
Maximum Campaign Success Rate	**30.0%** (Campaign B)	18.7%	CRITICAL
Annualized Loss Expectancy (ALE)	**$652.5M**	$285M	SEVERE
MFA Implementation ROI	**724,175%**	350%	EXCELLENT

3. Current State Financial Risk Profile:

- **Annualized Loss Expectancy (ALE):** $652,500,000
- **Expected Breaches:** 150 incidents (Derived from 600 compromised credentials × 25% probability)
- **Per-Employee Loss Exposure:** $652,500,000 / 2,000 = **$326,250**

DELIVERABLE 5: PROPOSED CONTROL INVESTMENT PORTFOLIO & ROI CALCULATIONS

CONTROL INITIATIVE	IMPLEMENTATION COST	RISK REDUCTION ($)	NET FINANCIAL BENEFIT ($)	ROI (%)	PRIORITY
MFA Implementation	$90,000	$651,847,500	$651,757,500	**724,175%**	1 (Immediate)
Enhanced Security Training	$100,000	$391,500,000	$391,400,000	**391,400%**	2 (Q1)
Advanced Email Filtering	$50,000	$130,500,000	$130,450,000	**260,800%**	3 (FY2024)

Expected Risk Posture After Full Control Suite

METRIC	CURRENT	WITH CONTROLS	REDUCTION
Credential Theft Rate	22.1%	2.2%	**90.0%**
Annual Loss Expectancy	$652.5M	$65.0M	**90.0%**
Breach Probability / Posture	93.75%	25.0%	**73.3%**

Total Investment Required: $490,000 | **Total Risk Reduction:** $587,500,000 | **Net Financial Benefit:** $587,010,000

★ BONUS CHALLENGE: Investment Break-Even Point

Question: If we invest $490,000 in security controls, how many breaches must we prevent to justify the cost?

Calculation:

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Break-Even = Total Investment / Cost per Data Breach
Break-Even = $490,000 / $4,350,000 = 0.1126 breaches
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Conclusion: Preventing just **0.113 of a single breach** (or preventing 1 data breach every 8.88 years) fully amortizes and justifies our entire $490,000 security investment portfolio.